



Evaluating the Sustainable Development Performance of G20 Economies Using an Integrated Decision-Making Framework

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Abstract: Assessing sustainable development performance is essential for understanding national progress. This study evaluates the sustainable development performance of G20 member countries (excluding the EU) using an integrated set of economic, social, and environmental indicators. Economic performance is captured by GDP per capita and the unemployment rate, while social performance is assessed through the education index, health expenditure, and income inequality (Gini coefficient). Environmental performance is represented by CO₂ emissions and the share of renewable energy in total energy consumption. Objective, data-driven weights were derived using the Criteria Importance Through Intercriteria Correlation (CRITIC) method, and countries were subsequently ranked using two multi-criteria decision-making (MCDM) approaches: Additive Ratio Assessment (ARAS) and Grey Relational Analysis (GRA). Correlation analysis (Spearman, Kendall, and Pearson) was conducted to examine the consistency and reliability of the rankings. The findings provide a comparative assessment of sustainability performance across G20 countries, highlighting relative strengths and weaknesses across economic, social, and environmental dimensions. The results offer a structured reference for decision-makers in engineering management and policy design in formulating evidence-based strategies.

Keywords: Sustainable development; G20 economies; Sustainability performance; Multi-criteria decision-making; Criteria Importance Through Intercriteria Correlation; Additive Ratio Assessment; Grey Relational Analysis

1 Introduction

Sustainable development has become a central concept, as it integrates economic growth, social justice, and environmental stewardship. The concept of sustainable development was formally introduced in the Brundtland Report, commonly referred to as *Our Common Future*, published in 1987 [1]. The report addressed development through economic, environmental, and social dimensions, highlighting that economic factors alone cannot fully support sustainable development [2]. This suggested framework is closely aligned with the United Nations Sustainable Development Goals to effectively assess development programs [3].

The G20 countries can be considered global decision-makers because they cover the world's economies, accounting for 85% of global GDP and 75% of international trade. This means that the G20 economies are leading players in global production, consumption, and emissions and therefore have the highest potential to drive sustainability consequences on a global scale. Nevertheless, notwithstanding their importance, significant variations in sustainability achievements exist across the economic, social, and environmental aspects among such countries [4]. Among global governance platforms, the G20 stands out due to its economic scale and influence on global sustainability outcomes. Therefore, it provides an appropriate context for comparative sustainability assessment.

Conventional evaluations of sustainability often focus on isolated measures such as GDP per capita or CO₂ emissions. A single metric fails to capture the multidimensional nature of sustainable development. A more holistic evaluation requires a methodology that considers many, often conflicting criteria simultaneously. Multi-criteria decision-making (MCDM) methods, such as the Analytic Hierarchy Process (AHP), Best–Worst Method (BWM), Elimination and Choice Expressing Reality (ELECTRE), Multi-Attribute Utility Theory (MAUT), Technique for

Order Preference by Similarity to Ideal Solution (TOPSIS), VIKOR, and Weighted Aggregated Sum Product Assessment (WASPAS), offer structured approaches for these evaluations, enabling decision-makers to prioritize alternatives according to multiple criteria [5–8].

A critical step in MCDM is the determination of the weights of criteria, which represents each factor's relative importance. A technique of weight assignment, Criteria Importance Through Intercriteria Correlation (CRITIC), calculates the weights based on data variability, minimizing subjectivity in the evaluation process [9]. The method enhances objectivity and reliability in ranking sustainability performance across G20 countries. In this study, the empirical analysis is conducted for G20 member countries (excluding the EU).

The primary objective of this research is to evaluate the overall development of G20 member countries (excluding the EU) by integrating economic, social, and environmental criteria using MCDM approaches. This study employs the CRITIC method to weigh criteria and simultaneously integrates Additive Ratio Assessment (ARAS) and Grey Relational Analysis (GRA) techniques to present a comprehensive sustainability performance evaluation. To examine the findings yielded by the methods, correlation analysis (Spearman rho, Kendall tau, and Pearson correlation) was performed, and the results are consistent and reliable. The study is a significant contribution in that it supports decision-makers by providing a perspective that highlights the strengths and weaknesses of national sustainability performance.

While this study primarily assesses sustainability outcomes across the G20 member countries (excluding the EU) included in the analysis, it also serves as a potential decision-support and benchmarking tool in engineering and systems management contexts. The proposed CRITIC–ARAS–GRA framework may contribute to system-level planning, resource allocation, and performance comparison in complex socio-technical systems. By integrating environmental, economic, and social indicators, the methodology provides a structured basis for prioritizing interventions and inform planning processes within engineering and industrial settings.

The main contribution of this study lies in its comprehensive evaluation approach that not only combines the latest MCDM techniques but also uses objective weighting for the assessment of sustainability performance of G20 member countries (excluding the EU) in economic, social and environmental sectors. The comprehensive approach provides a detailed insight of sustainability outcomes, which is highly valued by both policy makers and academic researchers. Besides, the dual application of ARAS and GRA, these two structurally different MCDM methods are used to verify the methodological robustness whereas most of the previous CRITIC based hybrid applications were employing only one ranking technique. While ARAS is based on additive utility aggregation, GRA is a method of measuring relational closeness to an ideal reference sequence under the conditions of uncertainty. The study, by implementing both methods with the same set of objective weights, provides a comparative validation procedure and also assesses the consistency of country rankings. This dual-structure evaluation enhances the methodological reliability as well as reduces the possibility of aggregation bias which is frequently the scenario in single-method sustainability assessments.

This study develops an integrated CRITIC–ARAS–GRA framework for multi-criteria sustainability assessment, providing a systematic method for weighting and ranking complex indicators.

- The framework may be applied as a benchmarking and decision-support tool in engineering systems, providing a structured basis for analyzing performance and informing resource allocation, risk considerations, and strategic planning.

- A comparative analysis of CRITIC–ARAS and CRITIC–GRA outcomes is presented, demonstrating methodological robustness and supporting informed decision-making in engineering management and policy contexts.

The remainder of this paper is organized as follows: Section 2 considers national sustainability performance from a systems perspective. Section 3 reviews literature on sustainable development performance, the use of MCDM, and engineering management studies. Section 4 presents the methodological framework and describes the CRITIC, ARAS, and GRA methods. Section 5 applies the methods to G20 member countries (excluding the EU) and reports the results. Finally, Section 6 presents the discussion and conclusions, including the main findings, policy implications, and directions for future research.

2 Systems View of National Sustainability Performance

In this study, we view countries as complex national sustainability systems composed of three tightly interconnected subsystems: economic, social, and environmental (Figure 1). The system boundary is set at the national level, where the inputs, processes, and outputs of each subsystem collectively determine the overall sustainability performance.

Economic subsystem: The economic subsystem refers to indicators such as GDP per capita, industrial productivity, and energy structure. Higher economic growth is often associated with improved social outcomes; for example, a rise in income may reduce unemployment levels while it may also, at the same time, raise inequality if no redistribution mechanisms are instituted.

Social subsystem: Reflects the status of health, education, and labor. Investments in health and education lead to a rise in productivity and social stability which ultimately contributes to economic growth.

Environmental subsystem: The environmental subsystem refers to energy transition, CO₂ emissions, and resource use. Changes in energy composition (e.g. introducing more renewables) significantly affect emission levels and influence economic performance in terms of costs and incentives for innovation.

The three subsystems continuously interact with each other, thereby creating feedback loops that influence national sustainability outcomes. For example, better education leads to a more skilled workforce which increases economic productivity while supporting social cohesion. Energy policy is an example of a factor that influences both emissions and economic costs, thereby demonstrating the interconnectedness of the system components.

Conceptualizing countries as systems enables the MCDM framework to capture interactions among subsystems and provide actionable insights for engineering and management decisions at the national level.

Figure 1 illustrates the schematic diagram of the systems view of national sustainability.

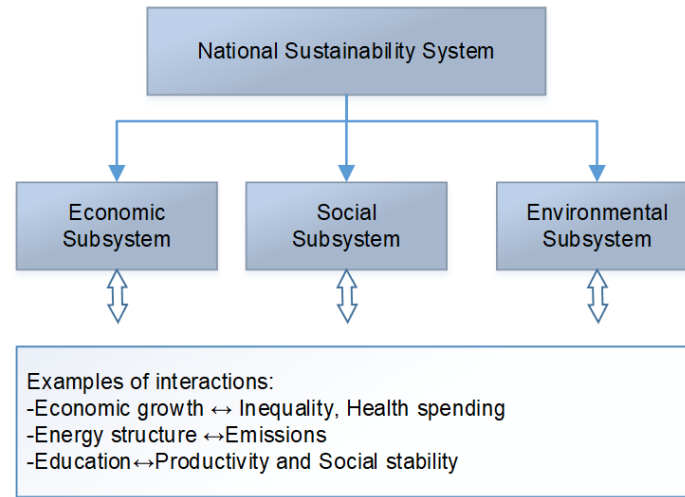


Figure 1. Schematic diagram of the national sustainability system

3 Literature Review

The G20 member countries (excluding the EU) included in the analysis comprise major economies from both developed and developing regions. Evaluating their sustainable development performance is essential due to the impact that they create on worldwide economic, environmental, and social factors. MCDM techniques have been applied for this assessment and have proven to be effective tools for enabling a comprehensive evaluation of different facets of sustainability. MCDM techniques support the assessment of conflicting criteria; therefore, they are highly suitable for the evaluation of sustainable development which encompasses economic, environmental, and social aspects. These techniques offer a framework for organized decision-making enabling policymakers to rank alternatives based on evaluations.

3.1 Sustainability Performance Assessment in Large-Scale Systems

The G20 economies are complex macro-level systems with a significant impact on global production networks, energy consumption patterns, changes in infrastructure investment, and environmental externalities. Therefore, assessing their sustainability performance becomes a complex task involving the evaluation of multidimensional systems, going far beyond mere policy adaptation. In the literature, various MCDM techniques have been used to assess sustainability dimensions in the G20 member countries (excluding the EU) and other large-scale contexts.

For instance, Gökgöz and Yalçın employed VIKOR and Combined Compromise Solution (CoCoSo) methods to analyze environmental and energy sustainability among G20 countries [10]. Altıntaş used Cost-Effectiveness Based Model (CEBM)-TOPSIS to assess marine ecosystem performance [11], while Geyik et al. [12] evaluated pandemic response efficiency using MCDM-based approaches. Öztaş and Öztaş [13] proposed a hybrid Logarithmic Percentage Change-driven Objective Weighting (LOPCOW)-Multi-Attributive Ideal-Real Comparative Analysis (MAIRCA) framework to measure innovation performance across G20 countries. Beyond G20-specific studies, Garcia-Bernabeu et al. [14] evaluated circular economy implementation in EU member states using MCDM-based approaches, providing transferable insights for large-scale economic systems. Reports such as McKinsey further emphasize that G20 countries account for approximately 31 gigatons of annual CO₂ emissions and must achieve significant reductions to meet net-zero targets [4].

Although these studies provide valuable comparative insights, most focus on specific sub-dimensions (e.g., energy efficiency, innovation, marine health, crisis management) rather than adopting an integrated systems-level perspective

on sustainability. Furthermore, the majority of these contributions are positioned within sustainability economics or public policy domains, with limited explicit linkage to engineering decision-support or systems-oriented analytical frameworks.

Recent engineering-oriented reviews suggest that sustainability evaluation problems increasingly benefit from structured analytical approaches capable of integrating multidimensional performance indicators in a consistent manner [15, 16]. These observations indicate the potential to interpret national sustainability assessment within a systems-oriented analytical perspective, rather than as a strictly engineering evaluation task.

3.2 Multi-Criteria Decision-Making Methods in Engineering Systems and Decision Analytics

MCDM methods have become fundamental tools in engineering systems analysis due to their ability to handle trade-offs among conflicting technical, economic, environmental, and operational criteria. In the field of engineering management, MCDM methods are among the most commonly used tools in allocating infrastructure, technology selection, renewable energy planning, industrial optimization and performance benchmarking. Based on thorough reviews, it has been shown that MCDM methods in engineering have experienced rapid growth, particularly in sustainable engineering, infrastructure evaluation, and industrial performance assessment [17, 18]. In the construction and infrastructure area, Villalba et al. [16] have shown the growth of the use of well-structured MCDM models for making retrofit and performance evaluation decisions. Likewise, the role of MCDM frameworks in renewable energy integration and energy management optimization has been highly recognized through energy system studies [19, 20].

From a methodological standpoint, recent literature emphasizes the importance of robustness and sensitivity assessment in MCDM applications. Nabavi et al. [21] show that ranking outcomes may vary depending on method selection and weighting structure, underlining the necessity of comparative evaluation. Likewise, Torkayesh et al. [22] document the evolution of modern ranking approaches and stress the importance of methodological cross-validation in decision analytics.

Moreover, hybrid decision support systems integrating advanced analytical tools and MCDM techniques are increasingly adopted in complex engineering environments, including intelligent transportation and autonomous systems planning [23]. These developments reflect a broader shift toward data-driven engineering decision analytics frameworks.

However, many empirical papers still depend on one-method ranking system or expert-based weighting approaches, such as AHP which may lead to bias and reliability issues. Objective weighting approaches like CRITIC provide a statistical basis for weighting by making use of data variability and inter-criteria correlation patterns to increase openness and logical consistency of analyses.

3.3 Engineering Management and Systems-Level Performance Evaluation

From an engineering management perspective, national sustainability performance may be viewed as a macro-scale systems evaluation problem involving interconnected subsystems (economic output, environmental impact, energy infrastructure, and social development). Such systems benefit from structured decision frameworks that aim to enhance transparency, consistency, and replicability in the evaluation process.

Engineering management literature generally emphasizes three important dimensions in performance evaluation frameworks:

1. Integration of multidimensional criteria structures;
2. Objective and data-driven weighting mechanisms;
3. Robust ranking validation and sensitivity testing.

However, comparative analysis of existing sustainability-oriented MCDM studies reveals several limitations. First, many analyses adopt fragmented indicator sets rather than constructing an integrated economic–social–environmental systems framework. Second, single ranking methods are frequently applied without cross-validation against structurally distinct alternatives. Third, subjective weighting dominance reduces methodological transparency and limits decision reliability.

Although CRITIC-based objective weighting has been applied in various engineering domains [17], its systematic integration with multiple ranking algorithms—such as ARAS and GRA—within a unified large-scale country-level performance evaluation framework remains limited. Furthermore, few studies explicitly assess ranking stability through correlation-based statistical validation, despite the methodological importance of robustness testing emphasized in recent decision analytics literature [21].

Accordingly, this study aims to contribute to the engineering management and systems-oriented decision-support literature by:

- Constructing an integrated multidimensional sustainability performance framework that is applicable to macro-scale evaluations.
- Employing objective CRITIC-based weighting to provide a transparent and data-driven basis for analysis.

- Comparing two structurally distinct MCDM ranking methods (ARAS and GRA) to offer a methodological perspective.

- Examining ranking consistency through statistical measures to support the robustness of the results.

By presenting performance results assessment within a structured, systems-oriented analytical framework rather than a purely policy-oriented perspective, this study establishes a potential linkage with engineering management and systems evaluation literature.

A comparative examination of the existing literature reveals three main patterns. Firstly, numerous studies are limited to a few specific sustainability sub-dimensions such as energy efficiency, marine health, innovation performance, or crisis management and they do not consider a fully integrated economic, social, environmental framework. Secondly, some studies rely on only one MCDM technique, thus not allowing methodological robustness and sensitivity testing. Thirdly, weighting methods often consist of subjective methods (e.g. AHP or expert, based methods) which can lead to bias in the evaluation process. In addition, hardly any papers explicitly assess the ranking stability through correlation analysis for alternative aggregations. Hence, this research fills the gap by (i) combining multidimensional sustainability metrics, (ii) using objective CRITIC-based weighting, (iii) contrasting two structurally distinct MCDM methods (ARAS and GRA), and (iv) statistically verifying the consistency of rankings. Such systematic unification ensures methodological soundness as well as higher policy relevance compared with previous fragmented approaches.

Therefore, the integration of these methods into one single analytical framework allows this research study to contribute to the literature. The application of two separate MCDM methodologies, such as ARAS and GRA, which are capable of dealing with uncertainty and incomplete information, significantly expands the scope of the sustainability performance evaluation. This kind of hybrid approach not only offers a more comprehensive view but also deepens our understanding of the sustainability features of G20 member countries (excluding the EU).

Many world-wide-scale institutions collaborate on economic, political, and financial issues. Among them, the G20 is one of the most extensive and influential bodies. The G20, as a forum for international economic cooperation, is similar to a few other world platforms in covering different facets of global governance. The G7, for instance, concentrates on developed countries' cooperation and usually establishes political priorities that the G20 later takes up [24]. On the other hand, BRICS as the group representing major emerging markets, advocates multipolarity and demands reforms in Western-dominated financial institutions [25]. The OECD, for its part, encourages policy harmonization and upholds international development standards based on evidence [26]. Besides, IMF and the World Bank offer financial instruments that help in maintaining global macroeconomic stability [27]. So, within this, a thorough and data-based assessment of sustainable development in G20 countries is to be done to expose their relative advantages and limitations.

4 Methods

4.1 Criteria Importance Through Intercriteria Correlation

The CRITIC method, originally developed by Diakoulaki et al. [28], is an objective weighting method used in MCDM problems. Unlike subjective methods that rely on expert judgment, CRITIC defines the weights of criteria based on the inherent features of the data, namely, the contrast intensity and the degree of conflict (correlation) among the criteria. It exhibits greater variation:

- It exhibits a greater amount of variation (i.e., it better discriminates among alternatives).
- It has lower correlation with other criteria (i.e., it provides unique information).

The CRITIC method follows these main steps:

- Build the Decision Matrix: Let $X = [x_{ij}]$ be the decision matrix and x_{ij} represents the performance of alternative i under criterion j .
- Normalize the Decision Matrix: To reduce the effect of dissimilar dimension units, the matrix is normalized using the min–max approach as given in Eq. (1).

$$r_{ij} = \begin{cases} \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}, & \text{for benefit criteria} \\ \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)}, & \text{for cost criteria} \end{cases} \quad (1)$$

- Compute the Standard Deviation of Each Criterion: The standard deviation σ_j is considered for each criterion j . A higher standard deviation indicates a greater level of contrast among options (as given in Eq. (2)).

$$\sigma_j = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (r_{ij} - \bar{r}_j)^2} \quad (2)$$

- Analyze the Correlation Matrix: The Pearson correlation coefficient r_{jk} between each pair of criteria j and k is calculated. This step evaluates the redundancy or interdependence among criteria.

- Calculate the Information Content (C_j) for Each Criterion: The amount of information (or the importance) provided by each criterion j is calculated using Eq. (3):

$$c_j = \sigma_j \sum_{k=1}^n (1 - r_{jk}) \quad (3)$$

This captures both the variability and the independence of the criterion.

- Decide the Weights: The weight w_j of each criterion is obtained by normalizing its information content as given in Eq. (4):

$$w_j = \frac{C_j}{\sum_{j=1}^n C_j} \quad (4)$$

4.2 Additive Ratio Assessment

ARAS is a MCDM method used to rank alternatives based on multiple criteria. It is simple and widely used in situations where alternatives need to be evaluated against both benefit and cost criteria. The method is based on comparing each alternative with an ideal alternative and computes a utility score to rank them.

Step 1: Construct the Decision Matrix: Assuming there are m alternatives and n criteria.

Build a decision matrix $X = [x_{ij}]$, where x_{ij} is the performance of alternative i under criterion j .

Step 2: Normalize the Decision Matrix: Normalize the values to make them dimensionless and comparable.

- For benefit criteria (higher values are preferred), as given in Eq. (5):

$$r_{ij} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}} \quad (5)$$

- For cost criteria (lower is better) as given in Eq. (6):

$$r_{ij} = \frac{\min(x_j)}{x_{ij}} \quad (6)$$

Step 3: Calculate the Weighted Normalized Matrix: Multiply the normalized values by their corresponding weights (Eqs. (7)–(8)):

$$v_{ij} = w_j r_{ij} \quad (7)$$

$$\sum_{j=1}^n w_j = 1 \quad (8)$$

Step 4: Determine the Optimal (Ideal) Alternative: Construct the ideal alternative A_0 by summing the best values of each criterion:

$$x_{0j} = \max(x_{ij}) \text{ for benefit criteria or } x_{0j} = \min(x_{ij}) \text{ for cost criteria}$$

Compute the overall utility for ideal alternative (Eq. (9)):

$$S_o = \sum_{i=1}^n v_{oj} \quad (9)$$

Step 5: Calculate the Utility Degree of Each Alternative: For each alternative i , compute the utility degree (Eq. (10)):

$$s_i = \sum_{j=1}^n v_{ij} \quad (10)$$

Then the relative utility (Eq. (11)):

$$K_i = \frac{S_i}{S_o} \quad (11)$$

A higher K_i indicates a better alternative.

Step 6: Rank the Alternatives: Rank alternatives according to K_i values. The alternative with the highest K_i is the best choice.

4.3 Grey Relational Analysis

GRA is an objective method used in MCDM processes to evaluate alternatives based on specific criteria. GRA provides an effective analytical tool, particularly in situations involving uncertainty and incomplete data. This method measures the relationship of a particular alternative with others and is used to determine the most suitable solution. The main advantage of GRA lies in its robustness against uncertainty within the dataset and its ability to minimize subjective judgments.

This method offers several advantages, including robustness under uncertainty and applicability with limited data, allowing easy calculation of grey relational coefficients, and not requiring the dataset to follow a specific distribution.

The foundation of GRA is to examine the relationship of each alternative with others based on certain criteria. This relationship is calculated using a grey relational coefficient, which indicates the degree of similarity between alternatives. A high coefficient signifies that alternatives are more similar to each other, while a low coefficient indicates greater dissimilarity. The steps of the GRA method are as follows:

1. Formation of the Decision Matrix

The decision matrix consists of m alternatives and n criteria. Here, y_{ij} represents the performance of the i -th alternative with respect to the j -th criterion.

2. Normalization of the Data

- If the data in the study are benefit-based—meaning that higher values are preferred (i.e., maximizing values)—they are normalized using the Eq. (12):

$$x_{ij} = \frac{y_{ij} - \text{Min} \{y_{ij}, i = 1, 2, \dots, m\}}{\text{Max} \{y_{ij}, i = 1, 2, \dots, m\} - \text{Min} \{y_{ij}, i = 1, 2, \dots, m\}} \quad (12)$$

$j = 1 \dots n$

- If the data in the study are cost-based—that is, if lower values are preferred (i.e., minimizing values)—they are normalized using the Eq. (13):

$$x_{ij} = \frac{\text{Max} \{y_{ij}, i = 1, 2, \dots, m\} - y_{ij}}{\text{Max} \{y_{ij}, i = 1, 2, \dots, m\} - \text{Min} \{y_{ij}, i = 1, 2, \dots, m\}} \quad (13)$$

$j = 1 \dots n$

- If the data in the study are neither purely benefit-based nor cost-based, and normalization needs to be performed relative to a specific optimal value, using the Eq. (14):

$$x_{ij} = 1 - \frac{|y_{ij} - y_j^*|}{\text{Max} \{ \text{Max} \{y_{ij}, i = 1, 2 \dots m\} - y_j^*, y_j^* - \text{Min} \{y_{ij}, i = 1, 2, \dots, m\} \}} \quad (14)$$

$i = 1, 2 \dots m$
 $j = 1, 2 \dots n$

3. Construction of the Grey Relational Coefficient Matrix

The grey relational coefficient is used to determine how close x_{ij} is to x_{0j} as given in Eq. (15) and Eq. (16):

$$\gamma(x_{0j}, x_{ij}) = \frac{\Delta_{\min} + \zeta \Delta_{\max}}{\Delta_{ij} + \zeta \Delta_{\max}} \quad (15)$$

$i = 1, 2 \dots m$
 $j = 1, 2 \dots n$

$$\begin{aligned} \Delta_{ij} &= |x_{0j} - x_{ij}| \\ \Delta_{\min} &= \text{Min} \{ \Delta_{ij}, i = 1, 2, \dots, m; j = 1, 2, \dots, n \} \\ \Delta_{\max} &= \text{Max} \{ \Delta_{ij}, i = 1, 2, \dots, m; j = 1, 2, \dots, n \} \quad \zeta(0 \leq \zeta \leq 1) \end{aligned} \quad (16)$$

where, $\zeta(0 \leq \zeta \leq 1)$ is defined as the distinguishability coefficient, and the smaller this value, the higher the distinguishability. Typically, ζ is assigned a value of 0.5 to reflect a moderate level of distinguishability.

4. Calculation of Grey Relational Grades

For criteria with equal importance, the first formula, Eq. (17) is used. However, for criteria with different levels of importance, the second formula, Eq. (18) is applied.

$$x_{ij} = 1 - \Gamma_{0i} = 1/m \sum_{j=1}^n \gamma_{0i}(j) \quad (17)$$

$$i = 1, 2, \dots, m$$

$$\Gamma_{0i} = \sum_{j=1}^n [w(j) * \gamma_{0i}(j)] \quad (18)$$

$$i = 1, 2, \dots, m$$

After calculating the grey relational grades, the similarity ratio among the series is evaluated with respect to the reference series, and the alternative with the highest grey relational grade is identified as the optimal solution.

5 Application of the Methods

This study focuses on G20 member countries (excluding the EU). Although the G20 includes the European Union, it is excluded from the analysis due to the requirement for consistent country-level data across all selected indicators. Comparable and complete data were not available for the European Union. Therefore, the empirical analysis is limited to the selected G20 countries. The analysis evaluates G20 countries across three dimensions: economic, social, and environmental indicators. These dimensions are selected to provide a comprehensive perspective on national performance in terms of economic growth, human development, and sustainability.

A. Indicator Selection

This study evaluates the G20 member countries (excluding the EU) by examining their performance in three critical domains—economic growth, social development, and environmental sustainability. These dimensions were selected to provide a comprehensive and multidimensional assessment of national development that extends beyond economic output alone.

The set of selected indicators was determined by four key considerations: (i) the completeness and reliability of data from all the G20 member countries (excluding the EU) for the whole time frame of the study, (ii) international comparability based on standardized definitions, (iii) the minimization of multicollinearity among the most closely related variables, and (iv) achieving sufficient representation in every dimension of sustainability while maintaining the simplicity of the model. Regarding the labour market, the unemployment rate was selected as a widely accepted macro-level proxy and provides an objective indicator. In addition, it is collected in a standardized manner and is available for all selected G20 countries. While different indicators such as informal employment or long-term unemployment could be used to obtain more comprehensive results, such variables are often plagued by incomplete coverage or definitional inconsistencies. Similarly, the share of renewable energy was selected as a key indicator of the energy transition, as it reflects structural changes in the national energy mix and is consistently reported across countries. Other indicators were considered; however, they either duplicate conceptually or bring about comparability restrictions.

GDP per capita and the unemployment rate were used as economic indicators to represent national prosperity, labour market productivity, and external competitiveness. Nevertheless, economic growth, by itself, is an insufficient indicator of the wellbeing of the people. This is the reason why social indicators were also used to measure human development outcomes. The Education Index, the share of GDP allocated on health, and the Gini coefficient are indicative of access to education, health financing, and the level of income inequality [29–31].

The inclusion of environmental metrics reflects the growing importance of sustainability in policy frameworks. CO₂ emissions per capita and the share of renewable energy in total consumption are widely accepted measures of environmental performance and progress, in energy transition, as defined by the Global Carbon Project [32] and the International Energy Agency (IEA) [33]. By integrating these three dimensions, this framework also aligning with the United Nations Sustainable Development Goals (SDGs), including SDG 8 (Decent Work and Economic Growth), SDG 10 (Reduced Inequalities), and SDG 13 (Climate Action), thereby enabling a balanced assessment of G20 member countries (excluding the EU)' progress toward sustainable and inclusive development.

B. Economic Indicators

Economic metrics reflect a nation's productive powers. They indicate how effectively resources are utilized to generate income, employment, and business.

- GDP per capita (Gross Domestic Product per capita): Economic indicators reflect a country's productive capacity.
- Unemployment rate: Reflecting the percentage of the labor force that is without work and actively seeking employment.

A high GDP per capita combined with low unemployment usually signifies strong economic performance, while a favorable trade surplus suggests that an economy is competitive.

C. Social Indicators

Social indicators reflect the standard of living, education, health, and equity among the population. They help measure progress and inclusion.

- Education Index: Part of the United Nations Development Programme (UNDP) Human Development Index; measures average years of schooling and expected years of education.
- Health Expenditure (% of GDP): Reflects the extent to which a country invests in its healthcare system relative to its economic performance.
- Income Inequality (Gini Coefficient): A measure of the inequality of income distribution in a country; 0 is perfect equality while 1 represents maximum inequality.

A higher education index and higher health spending augment capital; a lower Gini coefficient represents a more egalitarian society.

D. Environmental Indicators

Environmental indicators reflect how countries manage natural resources and ecological impacts as proof of their sustainability.

- CO₂ emissions per capita (tons): Reflects the amount of carbon dioxide emitted per person; lower values represent greener production and consumption of energy.
- Proportion of renewable energy within overall energy usage: Specifies what fraction of energy produced comes from renewable origins, such as solar, wind, and hydro. It contributes to the development of a low-carbon economy by reducing CO₂ emissions and increasing renewable energy use in accordance with the UN's (SDGs 7 and 13).

This study evaluates the selected countries based on three main dimensions—economic, social, and environmental indicators. These dimensions are chosen to provide a balanced understanding of each country's performance in terms of growth, human development, and sustainability. A sustainable economy reduces CO₂ emissions and increases renewable energy usage, aligning with the UN Sustainable Development Goals (SDGs 7 and 13).

Table 1 presents the evaluated factors and data sources. Table 2 presents the data obtained from the World Bank, the International Labour Organization (ILO), the World Health Organization (WHO), UNDP, and the IEA.

Table 1. Evaluated factors and the data sources

Dimension	Indicator	Description	Main Reference Source
Economic	GDP per capita	Measures income and productivity	[31]
	Unemployment rate	Reflects labor market strength	[34]
Social	Education index	Captures education quality	[29]
	Health expenditure (% of GDP)	Measures health investment	[30]
	Gini coefficient	Reflects income inequality	[31]
Environmental	CO ₂ emissions per capita	Reflects carbon footprint	[32]
	Renewable energy share	Reflects sustainability efforts	[33]

As part of the MCDM structure, G20 members are considered as decision alternatives. Only 19 alternatives have been taken into account in this study: Argentina (A1), Australia (A2), Brazil (A3), Canada (A4), China (A5), France (A6), Germany (A7), India (A8), Indonesia (A9), Italy (A10), Japan (A11), South Korea (A12), Mexico (A13), Russia (A14), Saudi Arabia (A15), South Africa (A16), Turkey (A17), the United Kingdom (A18), and the United States (A19).

The decision matrix consists of 19 alternatives (G20 countries) evaluated against seven criteria, including GDP per capita (C1), unemployment rate (C2), education index (C3), health expenditure (% of GDP) (C4), Gini coefficient (C5), CO₂ emissions per capita (C6), and renewable energy share (C7). Among these, C1, C3, C4, and C7 are classified as beneficial criteria, while C2, C5, and C6 are considered cost criteria. Table 2 presents the decision matrix used to evaluate the sustainable development performance of the selected countries.

E. Criteria Weighting Using the CRITIC Method

In this study, the CRITIC method was used to objectively assign weights to the sustainability criteria. Unlike subjective approaches such as AHP or Step-wise Weight Assessment Ratio Analysis (SWARA), CRITIC derives weights from the data's inherent characteristics—specifically, the standard deviation (representing contrast intensity) and the inter-criteria correlation (indicating redundancy). This data-driven technique ensures that criteria with

greater variability and weaker correlations with others receive higher weights, thereby having a stronger influence on the decision-making process.

The procedure begins with normalizing the decision matrix (Table 3) and constructing the inter criteria correlation matrix (Table 4). In Table 5, the standard deviations of the criteria are then calculated, and based on the derived information content values, the final criteria weights are computed. Table 5 summarizes these results, including the information content, standard deviations, weight values, and their corresponding rankings.

Table 2. Decision table

Country	GDP per Capita (USD)	Unemployment Rate (%)	Education Index	Health Expenditure (% of GDP)	Gini Index	CO ₂ Emissions (t/person)	Renewable Energy Share (%)
A1	10000	7.88	0.80	8.0	0.43	3.00	20
A2	60000	3.70	0.94	9.0	0.34	15.00	25
A3	15000	7.63	0.75	9.5	0.53	2.70	89
A4	70000	6.45	0.94	10.0	0.34	15.00	66
A5	15000	4.57	0.80	5.0	0.38	9.47	30
A6	45000	7.37	0.92	11.0	0.29	5.00	25
A7	55000	3.00	0.90	10.0	0.30	6.62	45
A8	2500	7.00	0.70	3.0	0.35	2.05	10
A9	4000	5.00	0.70	2.5	0.38	3.00	15
A10	40000	6.78	0.90	9.0	0.33	6.00	35
A11	45000	2.50	0.95	10.0	0.33	7.52	20
A12	35000	3.00	0.95	8.0	0.31	10.0	20
A13	10000	4.00	0.75	6.0	0.45	4.00	20
A14	12000	2.53	0.80	6.0	0.37	14.39	10
A15	25000	5.00	0.80	3.0	0.35	18.48	10
A16	5000	33.17	0.70	8.0	0.63	9.00	8
A17	10000	8.45	0.80	4.0	0.41	4.00	41.86
A18	55000	4.00	0.95	9.0	0.36	5.00	30
A19	80000	3.00	0.90	17.0	0.41	13.89	20

Note: The definitions of A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 3. Normalized decision table

	GDP per Capita (USD)	Unemployment Rate (%)	Education Index	Health Expenditure (% of GDP)	Gini Index	CO ₂ Emissions (t/person)	Renewable Energy Share (%)
A1	0.097	0.825	0.400	0.379	0.588	0.942	0.148
A2	0.742	0.961	0.960	0.448	0.853	0.212	0.210
A3	0.161	0.833	0.200	0.483	0.294	0.960	1.000
A4	0.871	0.871	0.960	0.517	0.853	0.212	0.716
A5	0.161	0.933	0.400	0.172	0.735	0.548	0.272
A6	0.548	0.841	0.880	0.586	1.000	0.820	0.210
A7	0.677	0.984	0.800	0.517	0.971	0.722	0.457
A8	0.000	0.853	0.000	0.034	0.824	1.000	0.025
A9	0.019	0.918	0.000	0.000	0.735	0.942	0.086
A10	0.484	0.860	0.800	0.448	0.882	0.760	0.333
A11	0.548	1.000	1.000	0.517	0.882	0.667	0.148
A12	0.419	0.984	1.000	0.379	0.941	0.516	0.148
A13	0.097	0.951	0.200	0.241	0.529	0.881	0.148
A14	0.123	0.999	0.400	0.241	0.765	0.249	0.025
A15	0.290	0.918	0.400	0.034	0.824	0.000	0.025
A16	0.032	0.000	0.000	0.379	0.000	0.577	0.000
A17	0.097	0.806	0.400	0.103	0.647	0.881	0.418
A18	0.677	0.951	1.000	0.448	0.794	0.820	0.272
A19	1.000	0.967	0.800	1.000	0.647	0.279	0.148

Note: The definitions of A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 4. Correlation of inter-criteria matrix

	C1	C2	C3	C4	C5	C6	C7
C1	1.000	0.342	0.862	0.783	0.469	-0.461	0.243
C2	0.342	1.000	0.443	0.043	0.753	-0.094	0.120
C3	0.862	0.443	1.000	0.625	0.638	-0.354	0.179
C4	0.783	0.043	0.625	1.000	0.029	-0.190	0.275
C5	0.469	0.753	0.638	0.029	1.000	-0.180	-0.087
C6	-0.461	-0.094	-0.354	-0.190	-0.180	1.000	0.163
C7	0.243	0.120	0.179	0.275	-0.087	0.163	1.000

Note: The definitions of C1–C7 are provided in Section 4.3 (Grey Relational Analysis).

Table 5. Information content, weights and rank of the criteria

	C1	C2	C3	C4	C5	C6	C7
C1	0.000	0.658	0.138	0.217	0.531	1.461	0.757
C2	0.658	0.000	0.557	0.957	0.247	1.094	0.880
C3	0.138	0.557	0.000	0.375	0.362	1.354	0.821
C4	0.217	0.957	0.375	0.000	0.971	1.190	0.725
C5	0.531	0.247	0.362	0.971	0.000	1.180	1.087
C6	1.461	1.094	1.354	1.190	1.180	0.000	0.837
C7	0.757	0.880	0.821	0.725	1.087	0.837	0.000
$\sum$ Column values	3.761	4.391	3.608	4.436	4.379	7.116	5.106
Std. deviation (σ)	0.317	0.219	0.373	0.241	0.244	0.308	0.253
C_j	1.192	0.962	1.345	1.068	1.068	2.189	1.293
W_j	0.131	0.106	0.148	0.117	0.117	0.240	0.142
rank	4	7	2	5	6	1	3

Note: The definitions of C1–C7 are provided in Section 4.3 (Grey Relational Analysis).

The results indicate that the criteria ranking is $C6 > C3 > C7 > C1 > C4 > C5 > C2$. The top criterion is CO₂ emissions (C6) while the least significant is the unemployment rate (C2). According to the CRITIC analysis, CO₂ emissions (C6) obtained a weight of 0.240 reflecting its considerable variability and low correlation with the other criteria. In contrast, the unemployment rate (C2) obtained the lowest weight (0.106), suggesting that it exhibits less variation and shares a moderate correlation with other factors. These derived weights will now serve as inputs for the subsequent MCDM analysis using the ARAS and GRA methods.

5.1 Criteria Importance Through Intercriteria Correlation Based Additive Ratio Assessment

The objective weights derived from the CRITIC method (Table 6) were utilized in the ARAS aggregation process.

The measurement data are presented in Table 6 (initial decision matrix), Table 7 (normalized and weighted normalized matrices), and Table 8 (relative utility results). Table 7 summarizes the normalized matrix, while Table 8 provides the calculated weighted normalized matrix.

The matrix is obtained by multiplying the normalized values with the CRITIC-based weights.

Based on the CRITIC-based ARAS approach, the alternatives are ranked as follows: $A19 > A4 > A7 > A18 > A11$, with the bottom three countries being A16, A9 and A15. According to Table 8, the ARAS results indicate that the United States obtained a score of 0.694, securing the first position. Canada (0.653), Germany (0.649), the United Kingdom (0.595), and Japan (0.583) follow, completing the top five positions.

Conversely, South Africa recorded the lowest score (0.226), indicating the weakest sustainable development performance among all the evaluated alternatives.

5.2 Criteria Importance Through Intercriteria Correlation Based Grey Relational Analysis

The decision matrix was created with the reference values in Table 9.

The normalized decision matrix is presented in Table 10. The normalization process is performed according to whether the criteria are benefit-based or cost-based.

In Table 11, the absolute value table was created by calculating the absolute differences between the values in the normalization matrices and the values of the reference series used in these matrices.

In the formula used to construct the grey relational coefficient matrix (Table 11), a value of 0.5 was applied for the separation coefficient, as this value is commonly recommended in the literature.

Firstly, grey relational degrees (Table 12) were calculated assuming that all variables (criteria) are equally important.

The determined values range from 0 to 1. The evaluated country with the performance score closest to 1 was considered the most ideal country, and rankings were made accordingly. When given equal importance, the resulting ranking was $A19 > A7 > A18 > A4 > A11$ (Table 12).

Table 13 presents the alternative rankings obtained using CRITIC-based weights, $A19 > A4 > A18 > A7 > A11$.

Table 6. Initial decision table

	Max C1	Min C2	Max C3	Max C4	Min C5	Min C6	Max C7
A1	10	7.88	0.8	8	0.43	3	20
A2	60	3.7	0.94	9	0.34	15	25
A3	15	7.63	0.75	9.5	0.53	2.7	89
A4	70	6.45	0.94	10	0.34	15	66
A5	15	4.57	0.8	5	0.38	9.47	30
A6	45	7.37	0.92	11	0.29	5	25
A7	55	3	0.9	10	0.3	6.62	45
A8	2.5	7	0.7	3	0.35	2.05	10
A9	4	5	0.7	2.5	0.38	3	15
A10	40	6.78	0.9	9	0.33	6	35
A11	45	2.5	0.95	10	0.33	7.52	20
A12	35	3	0.95	8	0.31	10	20
A13	10	4	0.75	6	0.45	4	20
A14	12	2.53	0.8	6	0.37	14.39	10
A15	25	5	0.8	3	0.35	18.48	10
A16	5	33.17	0.7	8	0.63	9	8
A17	10	8.45	0.8	4	0.41	4	41.86
A18	55	4	0.95	9	0.36	5	30
A19	80	3.5	0.9	17	0.41	13.89	20
Optimal value	80	2.5	0.95	17	0.29	2.05	89
sum	593.5		15.95	148	7.29	154.12	539.86

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 7. Normalized matrix

	C1	C2	C3	C4	C5	C6	C7
A1	0.017	0.031	0.050	0.054	0.045	0.095	0.037
A2	0.101	0.066	0.059	0.061	0.057	0.019	0.046
A3	0.025	0.032	0.047	0.064	0.037	0.105	0.165
A4	0.118	0.038	0.059	0.068	0.057	0.019	0.122
A5	0.025	0.053	0.050	0.034	0.051	0.030	0.056
A6	0.076	0.033	0.058	0.074	0.067	0.057	0.046
A7	0.093	0.081	0.056	0.068	0.065	0.043	0.083
A8	0.004	0.035	0.044	0.020	0.056	0.139	0.019
A9	0.007	0.048	0.044	0.017	0.051	0.095	0.028
A10	0.067	0.036	0.056	0.061	0.059	0.047	0.065
A11	0.076	0.097	0.060	0.068	0.059	0.038	0.037
A12	0.059	0.081	0.060	0.054	0.063	0.028	0.037
A13	0.017	0.061	0.047	0.041	0.043	0.071	0.037
A14	0.020	0.096	0.050	0.041	0.053	0.020	0.019
A15	0.042	0.048	0.050	0.020	0.056	0.015	0.019
A16	0.008	0.007	0.044	0.054	0.031	0.032	0.015
A17	0.017	0.029	0.050	0.027	0.048	0.071	0.078
A18	0.093	0.061	0.060	0.061	0.054	0.057	0.056
A19	0.135	0.069	0.056	0.115	0.048	0.020	0.037

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 8. Weighted normalized decision matrix

	C1	C2	C3	C4	C5	C6	C7	S_i	K_i	Rank
A1	0.004	0.005	0.007	0.007	0.005	0.011	0.004	0.043	0.386	11
A2	0.024	0.010	0.008	0.008	0.007	0.002	0.005	0.064	0.574	6
A3	0.006	0.005	0.007	0.008	0.004	0.012	0.017	0.060	0.536	8
A4	0.028	0.006	0.008	0.009	0.007	0.002	0.013	0.073	0.653	2
A5	0.006	0.008	0.007	0.004	0.006	0.004	0.006	0.041	0.366	15
A6	0.018	0.005	0.008	0.010	0.008	0.007	0.005	0.060	0.541	7
A7	0.022	0.012	0.008	0.009	0.008	0.005	0.009	0.072	0.649	3
A8	0.001	0.005	0.006	0.003	0.007	0.016	0.002	0.040	0.356	16
A9	0.002	0.007	0.006	0.002	0.006	0.011	0.003	0.037	0.334	18
A10	0.016	0.005	0.008	0.008	0.007	0.006	0.007	0.057	0.508	9
A11	0.018	0.014	0.008	0.009	0.007	0.004	0.004	0.065	0.583	5
A12	0.014	0.012	0.008	0.007	0.007	0.003	0.004	0.056	0.503	10
A13	0.004	0.009	0.007	0.005	0.005	0.008	0.004	0.042	0.379	12
A14	0.005	0.014	0.007	0.005	0.006	0.002	0.002	0.042	0.375	13
A15	0.010	0.007	0.007	0.003	0.007	0.002	0.002	0.037	0.334	17
A16	0.002	0.001	0.006	0.007	0.004	0.004	0.002	0.025	0.226	19
A17	0.004	0.004	0.007	0.004	0.006	0.008	0.008	0.041	0.367	14
A18	0.022	0.009	0.008	0.008	0.006	0.007	0.006	0.066	0.595	4
A19	0.032	0.010	0.008	0.015	0.006	0.002	0.004	0.077	0.694	1

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 9. Decision matrix

Alt.	C1	C2	C3	C4	C5	C6	C7
A1	10	7.88	0.8	8	0.43	3	20
A2	60	3.7	0.94	9	0.34	15	25
A3	15	7.63	0.75	9.5	0.53	2.7	89
A4	70	6.45	0.94	10	0.34	15	66
A5	15	4.57	0.8	5	0.38	9.47	30
A6	45	7.37	0.92	11	0.29	5	25
A7	55	3	0.9	10	0.3	6.62	45
A8	2.5	7	0.7	3	0.35	2.05	10
A9	4	5	0.7	2.5	0.38	3	15
A10	40	6.78	0.9	9	0.33	6	35
A11	45	2.5	0.95	10	0.33	7.52	20
A12	35	3	0.95	8	0.31	10	20
A13	10	4	0.75	6	0.45	4	20
A14	12	2.53	0.8	6	0.37	14.39	10
A15	25	5	0.8	3	0.35	18.48	10
A16	5	33.17	0.7	8	0.63	9	8
A17	10	8.45	0.8	4	0.41	4	41.86
A18	55	4	0.95	9	0.36	5	30
A19	80	3.5	0.9	17	0.41	13.89	20
Reference	80	2.5	0.95	17	0.29	2.05	89

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

In the initial phase of the GRA, all criteria are assumed to have equal importance, and the grey relational coefficients are calculated accordingly (Table 12). The results show that Alternative A19 achieved the highest grey relational grade, with A7, A18, A4, and A11 following behind whereas A14, A15, and A16 were identified as the lowest-performing alternatives.

In the subsequent phase, CRITIC-based weights are incorporated into the GRA to reflect the relative importance of each criterion. The weighted grey relational grades are presented in Table 13. Under this specification, the ranking slightly changes to A19 > A4 > A18 > A7 > A11.

A19 maintains its leading position with the highest relational score (0.766), whereas A16 remains the lowest-ranked alternative. The limited variation between the equal-weighted and weighted results suggests that the ranking

structure is relatively stable and not highly sensitive to the weighting scheme.

When the results of equal-weighted GRA are compared with those from the CRITIC-weighted GRA, the ranking pattern remains largely unchanged, except for minor position changes among the top alternatives. Such consistency between the two analyses indicates the robustness and reliability of the results, as well as their limited sensitivity to variations in the weighting method.

Table 10. The normalized decision matrix

Alt.	C1	C2	C3	C4	C5	C6	C7
A1	0.097	0.825	0.400	0.379	0.588	0.942	0.148
A2	0.742	0.961	0.960	0.448	0.853	0.212	0.210
A3	0.161	0.833	0.200	0.483	0.294	0.960	1.000
A4	0.871	0.871	0.960	0.517	0.853	0.212	0.716
A5	0.161	0.933	0.400	0.172	0.735	0.548	0.272
A6	0.548	0.841	0.880	0.586	1.000	0.820	0.210
A7	0.677	0.984	0.800	0.517	0.971	0.722	0.457
A8	0.000	0.853	0.000	0.034	0.824	1.000	0.025
A9	0.019	0.918	0.000	0.000	0.735	0.942	0.086
A10	0.484	0.860	0.800	0.448	0.882	0.760	0.333
A11	0.548	1.000	1.000	0.517	0.882	0.667	0.148
A12	0.419	0.984	1.000	0.379	0.941	0.516	0.148
A13	0.097	0.951	0.200	0.241	0.529	0.881	0.148
A14	0.123	0.999	0.400	0.241	0.765	0.249	0.025
A15	0.290	0.918	0.400	0.034	0.824	0.000	0.025
A16	0.032	0.000	0.000	0.379	0.000	0.577	0.000
A17	0.097	0.806	0.400	0.103	0.647	0.881	0.418
A18	0.677	0.951	1.000	0.448	0.794	0.820	0.272
A19	1.000	0.967	0.800	1.000	0.647	0.279	0.148

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 11. Absolute value table

Alt.	C1	C2	C3	C4	C5	C6	C7
A1	0.903	0.175	0.600	0.620	0.411	0.057	0.851
A2	0.258	0.039	0.040	0.551	0.147	0.788	0.790
A3	0.838	0.167	0.800	0.517	0.705	0.039	0.000
A4	0.129	0.128	0.040	0.482	0.147	0.788	0.284
A5	0.838	0.067	0.600	0.827	0.264	0.451	0.728
A6	0.451	0.158	0.120	0.413	0.000	0.179	0.790
A7	0.322	0.016	0.200	0.482	0.029	0.278	0.543
A8	1.000	0.146	1.000	0.965	0.176	0.000	0.975
A9	0.980	0.081	1.000	1.000	0.264	0.057	0.913
A10	0.516	0.139	0.200	0.551	0.117	0.240	0.666
A11	0.451	0.000	0.000	0.482	0.117	0.332	0.851
A12	0.580	0.016	0.000	0.620	0.058	0.483	0.851
A13	0.903	0.048	0.800	0.758	0.470	0.118	0.851
A14	0.877	0.001	0.600	0.758	0.235	0.751	0.975
A15	0.709	0.081	0.600	0.965	0.176	1.000	0.975
A16	0.967	1.000	1.000	0.620	1.000	0.423	1.000
A17	0.903	0.194	0.600	0.896	0.352	0.118	0.582
A18	0.322	0.048	0.000	0.551	0.205	0.179	0.728
A19	0.000	0.032	0.200	0.000	0.352	0.720	0.851
$\Delta_{\max}$	1.000						
$\Delta_{\min}$	0.000						
ζ	0.5						

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 12. Gray relational coefficient matrix (equal importance levels)

Alt.	C1	C2	C3	C4	C5	C6	C7	r_{0i}	rank
A1	0.356	0.740	0.455	0.446	0.548	0.896	0.370	0.545	12
A2	0.660	0.927	0.926	0.475	0.773	0.388	0.388	0.648	8
A3	0.373	0.749	0.385	0.492	0.415	0.927	1.000	0.620	10
A4	0.795	0.795	0.926	0.509	0.773	0.388	0.638	0.689	4
A5	0.373	0.881	0.455	0.377	0.654	0.525	0.407	0.525	16
A6	0.525	0.759	0.806	0.547	1.000	0.736	0.388	0.680	6
A7	0.608	0.968	0.714	0.509	0.944	0.643	0.479	0.695	2
A8	0.333	0.773	0.333	0.341	0.739	1.000	0.339	0.551	11
A9	0.338	0.860	0.333	0.333	0.654	0.896	0.354	0.538	13
A10	0.492	0.782	0.714	0.475	0.810	0.675	0.429	0.625	9
A11	0.525	1.000	1.000	0.509	0.810	0.600	0.370	0.688	5
A12	0.463	0.968	1.000	0.446	0.895	0.508	0.370	0.664	7
A13	0.356	0.911	0.385	0.397	0.515	0.808	0.370	0.535	15
A14	0.363	0.998	0.455	0.397	0.680	0.400	0.339	0.519	17
A15	0.413	0.860	0.455	0.341	0.739	0.333	0.339	0.497	18
A16	0.341	0.333	0.333	0.446	0.333	0.542	0.333	0.380	19
A17	0.356	0.720	0.455	0.358	0.586	0.808	0.462	0.535	14
A18	0.608	0.911	1.000	0.475	0.708	0.736	0.407	0.692	3
A19	1.000	0.939	0.714	1.000	0.586	0.410	0.370	0.717	1

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

Table 13. Gray relational degrees (CRITIC based weights)

	C1	C2	C3	C4	C5	C6	C7	r_{0i}	rank
A1	0.086	0.109	0.064	0.058	0.064	0.105	0.039	0.526	11
A2	0.158	0.137	0.131	0.062	0.091	0.045	0.041	0.666	7
A3	0.090	0.111	0.055	0.064	0.049	0.109	0.106	0.582	10
A4	0.191	0.117	0.131	0.066	0.091	0.045	0.067	0.709	2
A5	0.090	0.130	0.064	0.049	0.077	0.062	0.043	0.514	16
A6	0.126	0.112	0.114	0.072	0.117	0.086	0.041	0.668	6
A7	0.146	0.143	0.101	0.066	0.111	0.075	0.051	0.693	4
A8	0.080	0.114	0.047	0.045	0.087	0.117	0.036	0.526	12
A9	0.081	0.127	0.047	0.044	0.077	0.105	0.037	0.518	14
A10	0.118	0.115	0.101	0.062	0.095	0.079	0.045	0.616	9
A11	0.126	0.148	0.142	0.066	0.095	0.070	0.039	0.686	5
A12	0.111	0.143	0.142	0.058	0.105	0.060	0.039	0.658	8
A13	0.086	0.134	0.055	0.052	0.060	0.095	0.039	0.520	13
A14	0.087	0.147	0.064	0.052	0.080	0.047	0.036	0.513	17
A15	0.099	0.127	0.064	0.045	0.087	0.039	0.036	0.497	18
A16	0.082	0.049	0.047	0.058	0.039	0.063	0.035	0.374	19
A17	0.086	0.106	0.064	0.047	0.069	0.095	0.049	0.515	15
A18	0.146	0.134	0.142	0.062	0.083	0.086	0.043	0.696	3
A19	0.240	0.139	0.101	0.131	0.069	0.048	0.039	0.766	1

Note: The definitions of C1–C7 and A1–A19 are provided in Section 4.3 (Grey Relational Analysis).

5.3 Discussion: Comparison of Criteria Importance Through Intercriteria Correlation-Based Additive Ratio Assessment and Grey Relational Analysis Rankings

Table 14 shows a comparison between the rankings derived from the CRITIC-based ARAS and CRITIC-based GRA approaches. Both techniques seek to assess the development performance of nations by utilizing the objective weights obtained through the CRITIC method.

Carbon dioxide emissions are given high weight not only because of their high statistical variability, but also because of their inherent importance to countries. The production structures, energy systems, and environmental policy frameworks of the assessed countries differ extensively. For instance, developed countries have moved towards service-oriented economies and have improved their energy efficiency, which has resulted in more or less stable or even declining emission trends. In contrast, a number of emerging economies are still heavily dependent on the

carbon-intensive industrial production and fossil fuel consumption for their economic growth. Hence, the high ranking of CO₂ emissions in the CRITIC weighting method conveys that the carbon aspect still constitutes the main differentiating factor when making cross-country sustainability comparisons. This finding may also be interpreted as an indication of the importance of decarbonization in shaping cross-country sustainability differences.

Table 14. Comparison of the methods

	CRITIC Based ARAS		CRITIC Based GRA	
	Alt.	Rank	Alt.	Rank
USA	A19	1	A19	1
Canada	A4	2	A4	2
Germany	A7	3	A18	3
United Kingdom	A18	4	A7	4
Japan	A11	5	A11	5
Australia	A2	6	A6	6
France	A6	7	A2	7
Brazil	A3	8	A12	8
Italy	A10	9	A10	9
South Korea	A12	10	A3	10
Argentina	A1	11	A1	11
Mexico	A13	12	A8	12
Russia	A14	13	A13	13
Turkey	A17	14	A9	14
China	A5	15	A17	15
India	A8	16	A5	16
Saudi Arabia	A15	17	A14	17
Indonesia	A9	18	A15	18
South Africa	A16	19	A16	19

Note: The definitions of A1–A19 are provided in Section 4.3 (Grey Relational Analysis). CRITIC = Criteria Importance Through Intercriteria Correlation; ARAS = Additive Ratio Assessment; GRA = Grey Relational Analysis.

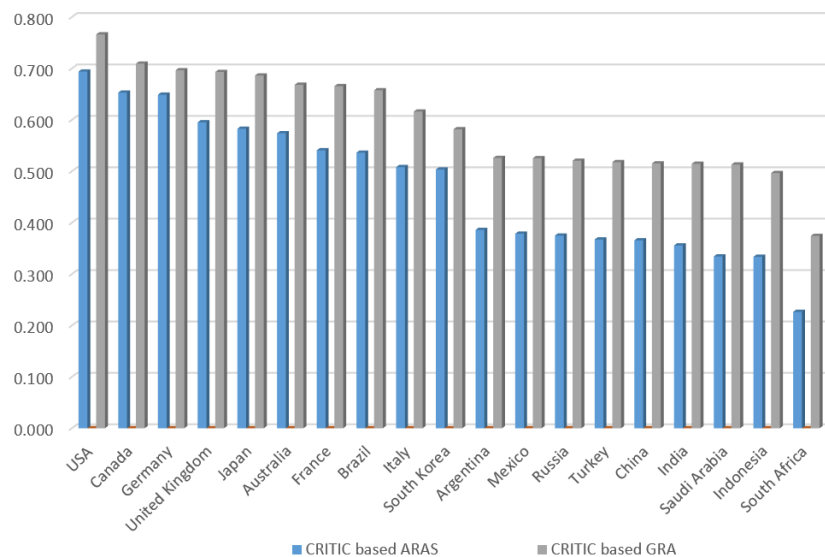


Figure 2. Bar chart of the methods

The results indicate that the top two countries, the United States (A19) and Canada (A4), are consistently ranked highest by both methods, suggesting a high level of agreement among the leading countries in terms of sustainability performance.

Similarly, South Africa (A16) ranks last, being considered the lowest performing country in both methods. However, only minor differences are observed among mid-ranked countries. For example, Germany (A7) and the United Kingdom (A18) exchange their rankings when comparing the ARAS and GRA methods with Germany

staying in place in ARAS but going down to fourth in GRA and the UK coming up from fourth in ARAS to third in GRA. Also, a few moderate differences are specified, such as France (A6) and Australia (A2) changing their 7th ranks between the two methods.

These minor differences are expected due to the distinct aggregation mechanisms of ARAS and GRA. Although minor differences exist, the overall ranking patterns remain highly consistent, particularly for the highest- and lowest-ranked countries. Overall, both methods produce highly similar ranking patterns, with differences limited to mid-level positional shifts.

Figure 2 shows the performance result scores of the G20 member countries (excluding the EU) computed by the two methods CRITIC-based ARAS and CRITIC-based GRA. However, the GRA method generally yields higher performance scores for the majority of the countries compared to ARAS method. Therefore, the difference between the two methods lies in the fact that, although countries are ranked similarly, they give different score magnitudes, reflecting a methodological effect stemming from the summation logic used.

The fact that the United States remains in the lead with both methods used points to a robust ranking that does not depend on the method. Also, Canada, Germany, the United Kingdom, and Japan rank among the top-performing countries, indicating that they possess structural sustainability advantages that are not highly sensitive to the aggregation technique. This stability reinforces the validity of the cross-method comparison. On the contrary, lower-ranked countries such as Indonesia, Saudi Arabia, and South Africa consistently remain at the bottom in both models. Their consistent position suggests that their weaknesses are not limited to particular indicators. Significantly, South Africa's relatively low ARAS score shows that aggregating utility additively may penalize imbalance across dimensions more substantially across the dimensions than the relational structure of GRA.

In sum, the bar chart indicates that the differentiation of countries within the considered countries is significant and largely reflects the emphasis on structural sustainability rather than the choice of the different methodologies.

Figure 3 illustrates a sustainability performance multidimensional view in a radar framework. The more extensive coverage limited by the GRA path validates the regular upward scoring trend seen in Figure 3; nevertheless, the almost identical geometry of both patterns shows a very strong structural agreement between the two methods.

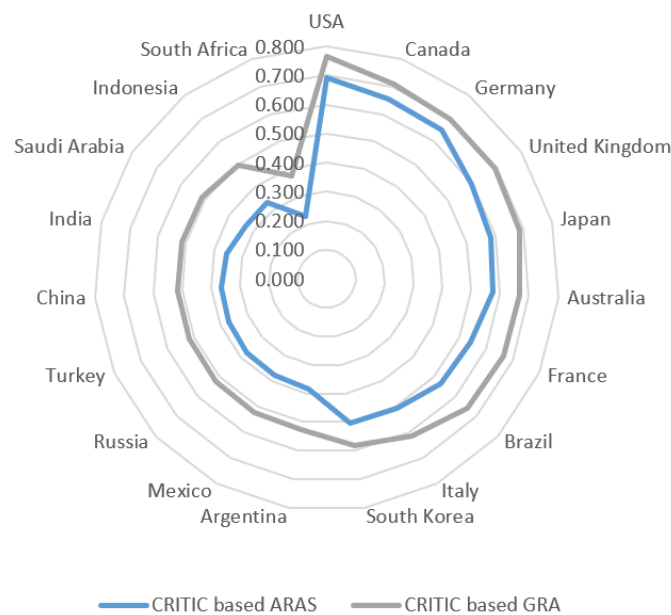


Figure 3. Radar chart of the methods

On the other hand, countries such as the United States, Canada, and Germany extend across multiple axes, thus indicating that they have well-balanced sustainability profiles instead of having just a few strong features. The fact that they are spreading outwards reflects their multidimensional consistency, which is very important for composite sustainability outcomes.

Among other countries, South Africa, Indonesia, and Saudi Arabia which are located near the center, have limited radar coverage, indicating cumulative weaknesses in several indicators rather than being only characterized by one-dimension deficiencies. This indicates that the countries' commitment to the sustainability principles is fundamentally limited rather than a short-term departure from good performance. It is worth noting that the almost identical shapes of the ARAS and GRA polygons reveal that the choice of method results in more differences in score size than in the inherent pattern of the rankings. This graphical agreement also supports the next correlation analysis

and strengthens the result's validity. This indicates that, in addition to confirming ranking similarity, the radar plot also reveals the leading economies' multidimensional harmony and the structural grouping of the lower-performing countries within the G20 sustainability assessment framework.

The visual consistency across both figures further suggests that sustainability performance within the 19 G20 is primarily shaped by structural economic and energy-system characteristics rather than methodological specification, reinforcing the policy relevance of the findings.

5.3.1 Indicative policy considerations

The analysis above provides actionable insights for country-specific sustainability policies:

1. High-performing nations (USA, Canada, Germany, UK, Japan)

1) Strengths: Balanced performance across environmental, social, and economic indicators; effective energy transition.

2) Policy Recommendations:

- Maintain leadership by accelerating clean technology adoption and low-carbon innovation.
- Leverage expertise to support knowledge transfer initiatives to emerging economies.

2. Low-performing nations (Indonesia, Saudi Arabia, South Africa)

1) Weaknesses: Challenges in energy transition, labor conditions, and CO₂ emissions.

2) Policy Recommendations:

- Energy: Increase investment in renewable energy, reduce fossil fuel dependency.
- Labor: Policymakers may consider implementing targeted job creation programs and vocational training to reduce unemployment.

• Environment: May consider regulatory measures such as carbon pricing or emission reduction incentives to curb CO₂ output.

3. Medium-performing nations (Brazil, India, Turkey)

1) Weaknesses: Specific domains show gaps, e.g., renewable energy share or social indicators.

2) Policy Recommendations:

- Prioritize interventions in underperforming areas, such as targeted incentives for renewable energy projects or social protection programs to improve labor outcomes.

The discrepancies between ARAS and GRA results disclose the areas where countries show imbalanced development in different dimensions, thereby offering policymakers a tool for intervention planning based on the evidence. The utilization of the CRITIC-ARAS-GRA framework together not only ensures agreement in the ranking of countries but also highlights the extent of performance differences. This enables customized, data-driven policy recommendations, thereby increasing the implementation value of sustainability evaluations of the 19 G20.

The results may also be interpreted from an engineering decision-support perspective. Higher-ranked countries can serve as indicative reference points for comparative system performance, while mid-ranked countries highlight areas where improvements may be considered based on relatively lower-performing criteria.

From an engineering management standpoint, the findings may provide a useful basis for benchmarking in energy transition contexts, exploring subsystem-level interventions, and informing portfolio-level investment considerations. Table 15 presents a concise summary of these decision-oriented interpretations.

Table 15. Decision interpretation of results

Rank Group	Key Weak Criteria	Suggested Decision Action
Top-ranked	N/A (benchmark level)	Use as reference benchmarks and best-practice targets for overall system performance. Maintain strengths while monitoring for early signs of decline.
Mid-ranked	Renewable energy share; healthcare spending	Apply targeted improvement levers in underperforming subsystems. Prioritize efficiency gains and policy adjustments that can yield medium-term impact.
Bottom-ranked	CO ₂ emissions; education; labour conditions	Deploy high-priority interventions with concentrated resource allocation. Consider structural reforms, regulatory action, and external support mechanisms where needed.

As shown in Table 16, the correlation analysis indicates a very high degree of agreement between the rankings produced by the CRITIC-based ARAS and CRITIC-based GRA methods. The monotonic relationship between the

two sets of ranks is almost perfect, according to Spearman's rho ($\rho = 0.944$). Kendall's tau ($\tau = 0.825$) highlights the rankings' stability by confirming a high degree of concordance with only a few rank inversions. A near-linear relationship between the rankings is further supported by Pearson's correlation ($r = 0.944$).

Table 16. Correlation analysis between Criteria Importance Through Intercriteria Correlation (CRITIC)-based Additive Ratio Assessment (ARAS) and CRITIC-based Grey Relational Analysis (GRA)

Correlation Method	Correlation Coefficient	Interpretation
Spearman ρ	0.944	Very strong monotonic relationship; ranks are highly consistent
Kendall τ	0.825	High concordance; only a few rank inversions
Pearson r	0.944	Very strong linear relationship between ranks

Together, these findings suggest that both MCDM techniques produce highly consistent rankings within the defined analytical framework. This indicates that the relative ordering of alternatives is not strongly affected by the choice of method. However, the results should be interpreted within the scope of the selected indicators and weighting approach.

5.4 Managerial and System-Level Insights

In addition to ranking countries by their level of sustainable development, the results reveal patterns that can inform managerial and system-level decision-making:

1. High GDP per capita and a high education index correlate with superior sustainability performance.

Implication: Countries with strong economic structures and human capital can serve as benchmarks in system-level planning, with policies focused on sustaining education and productivity gains.

2. Social inequality (Gini index) can offset good economic performance.

Implication: The presence of social inequality (measured by the Gini index) may require high-income countries to implement substantial social interventions; reducing the social divide is essential for improving overall system performance.

Table 17. Decision interpretation of key patterns

Pattern/Observation	Implication for System Planning	Suggested Action
High GDP per capita & high education index → higher overall score	Benchmarking for best practices; leverage strong human and economic capital	Focus on sustaining education, innovation, and productivity policies
High social inequality (Gini) offsets good economic performance	Inequality can undermine sustainability even in wealthy nations	Targeted social programs and policies to reduce inequality
CO ₂ emissions carry highest weight in assessment	Emissions are key driver of overall sustainability	Prioritize emission reduction, energy transition, and low-carbon investments
Mid-ranked countries often show low renewable energy share	Energy transition is a leverage point	Invest in renewable infrastructure and efficiency measures
Bottom-ranked countries have weak health and labor indicators	Social subsystem constraints limit performance	Prioritize healthcare, labor market improvements, and social protection policies

3. CO₂ emissions carry a high weight in the integrated assessment.

Implication: Policies targeting emission reductions and energy transition are likely to have a strong positive impact on sustainability performance, providing a solid basis for decision-making.

These patterns demonstrate that the integrated CRITIC–ARAS–GRA framework is not only an effective tool for analyzing economic, social, and environmental subsystems, but also a practical basis for resource allocation, intervention prioritization, and strategic planning. From a managerial and system-planning perspective, engineering

managers may use the performance scores as references to support decision-making and prioritize areas for further analysis. The key implications derived from these patterns are summarized in Table 17.

5.4.1 Transferability of the CRITIC–ARAS–GRA Framework

The proposed CRITIC–ARAS–GRA framework is not limited to national sustainability assessment and can be adapted to various engineering systems and decision-support contexts. It provides a structured approach for performance benchmarking, resource prioritization, and informed managerial decision-making. For instance, in infrastructure management, alternatives might be different projects, and criteria may include cost, time, risk, safety, emissions, and reliability. In the case of supply chain or energy portfolio assessment, alternatives are, for example, different operational strategies or asset portfolios, while criteria may include performance, risk, environmental impact, and economic efficiency.

Moreover, the CRITIC method determines weights objectively based on data, while the ARAS and GRA methods provide ranking results and relative closeness measures; therefore, the integrated approach is suitable for decision-making scenarios involving mixed benefit and cost criteria, as well as incomplete or uncertain data. This transferability underscores the framework's potential as a structured tool for engineering decision-support and macro-scale systems performance evaluation.

6 Conclusion and Future Directions

This study provides a comparative assessment of the sustainability performance of the G20 member countries (excluding the EU), integrating economic, social, and environmental dimensions. The analysis excludes the European Union and is limited to the 19 sovereign G20 member countries included in the study. The selected indicators include GDP per capita, unemployment rate, education index, healthcare expenditure, income inequality (Gini index), CO₂ emissions, and the share of renewable energy. The combination of CRITIC method for weighting and ARAS and GRA for ranking enables the analysis to account for both criteria variability and interrelationships, while providing a structured and data-driven benchmarking framework. This study aims to systematically analyze the development performance of selected countries using an MCDM approach. The results highlight relative strengths and weaknesses under the defined set of indicators and weights, but do not establish causal relationships or provide definitive policy prescriptions. For example, the United States and Canada ranked among the highest-performing countries, whereas South Africa consistently appeared at the bottom of the rankings. The results reflect the combined effects of economic, social, and environmental indicators within the defined framework and do not imply absolute superiority or inferiority.

Slight differences in rankings between ARAS and GRA methods illustrate the sensitivity of the MCDM approach to the specific aggregation technique employed. Nevertheless, these variations also underscore the robustness of using multiple methods for comparative benchmarking. The findings should be interpreted as indicative of relative performance under the selected criteria rather than as a comprehensive assessment of sustainability.

Overall, the study provides a quantitative framework that can serve as a reference for decision support, system-level planning, and further research. The proposed methodology may be adapted in future studies for scenario analysis or dynamic modeling of sustainability outcomes, but conclusions should remain confined to the scope of the indicators and weighting approach used in this research.

The CRITIC–ARAS–GRA framework may support engineering managers and system planners by providing a structured approach to prioritize improvement areas and evaluate trade-offs across economic, social, and environmental dimensions. The resulting country rankings can serve as a benchmarking reference, potentially informing goal-setting and highlighting best practices within complex socio-technical systems.

While the framework has potential applications in policy evaluation, infrastructure planning, and system performance assessment, it should be regarded as a decision-support tool rather than definitive guidance for managerial or engineering interventions. The results underscore the relative importance of selected criteria and the comparative performance of countries, offering insights to guide further analysis or strategic planning without overstating the study's scope.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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